

Supplementary Materials for “OmniGene-4: A Unified Bio-Language MoE Model with Router-Level Interpretability” (covering v3 routing analysis and v4-v5 SFT extensions)

Liang Wang
School of Artificial Intelligence and Automation
Huazhong University of Science and Technology

May 2026

Contents

1	Supplementary Tables	3
1.1	Supplementary Table S1: Full Per-Layer JS Divergence	3
1.2	Supplementary Table S2: Bootstrap Confidence Intervals	4
1.3	Supplementary Table S3: Format-Matched Routing Control	4
1.4	Supplementary Table S4: Head-to-Head Comparison Against Classical and PLM Baselines	4
1.5	Supplementary Table S8: Case-Study Pair Details	5
1.6	Supplementary Table S9: Layer-12 Top-3 Routed Experts per Case	5
2	Supplementary Figures	6
2.1	Supplementary Figure S1: Per-Layer JS Curves	6
2.2	Supplementary Figure S2: Expert Activation Heatmap	6
2.3	Supplementary Figure S3: Bootstrap Distribution	7
3	Supplementary Methods	8
3.1	Detailed Training Hyperparameters	8
3.1.1	Continued Pretraining (CPT)	8
3.1.2	Supervised Fine-Tuning (SFT)	9
3.2	Data Mixture Composition	9
3.2.1	CPT Data Mixture	9
3.2.2	SFT Data Mixture	10
3.3	Routing Collection Protocol	10
3.3.1	Forward Hook Implementation	10
3.3.2	Bootstrap Resampling Protocol	10
4	Supplementary Results	12
4.1	Token-Level Expert Specialization	12
4.2	Per-Task Entropy Analysis	12
4.3	Supplementary Table S5: Multi-task Generation Evaluation (v5)	12

4.4	Supplementary Table S6: Per-residue Classification Head Accuracy (v5)	13
4.5	Supplementary Table S7: Standard / Remote Homology Updated Numbers	13
5	Code Availability	14
5.1	Key Scripts	14
5.2	Data Files	14
5.3	Model Weights	15

1 Supplementary Tables

1.1 Supplementary Table S1: Full Per-Layer JS Divergence

Table 1: Layer-by-layer Jensen-Shannon divergence across training stages. Values are mean pairwise JS across all task pairs.

Layer	Baseline	CPT	v3 (CPT+SFT)	Δ CPT
0	0.160	0.164	0.164	+0.004
1	0.136	0.113	0.113	-0.023
2	0.172	0.164	0.164	-0.008
3	0.151	0.134	0.134	-0.017
4	0.119	0.113	0.113	-0.006
5	0.159	0.160	0.160	+0.001
6	0.131	0.111	0.111	-0.020
7	0.129	0.123	0.123	-0.006
8	0.134	0.126	0.126	-0.008
9	0.174	0.153	0.153	-0.021
10	0.197	0.159	0.159	-0.038
11	0.160	0.143	0.143	-0.017
12	0.182	0.149	0.149	-0.033
13	0.143	0.127	0.127	-0.016
14	0.157	0.137	0.137	-0.020
15	0.198	0.170	0.170	-0.028
16	0.139	0.121	0.121	-0.018
17	0.184	0.166	0.166	-0.018
18	0.148	0.134	0.134	-0.014
19	0.124	0.123	0.123	-0.001
20	0.140	0.130	0.130	-0.010
21	0.136	0.119	0.119	-0.017
22	0.162	0.149	0.149	-0.013
23	0.157	0.138	0.138	-0.019
24	0.141	0.122	0.122	-0.019
25	0.180	0.149	0.149	-0.031
26	0.161	0.152	0.152	-0.009
27	0.178	0.172	0.172	-0.006
28	0.217	0.197	0.197	-0.020
29	0.240	0.222	0.222	-0.018

1.2 Supplementary Table S2: Bootstrap Confidence Intervals

Table 2: Bootstrap 95% confidence intervals for routing metrics (1,000 iterations).

Metric	Mean	95% CI Lower	95% CI Upper
Baseline JS	0.344	0.344	0.344
CPT JS	0.440	0.440	0.441
v3 JS	0.444	0.444	0.444
Δ CPT	0.096	0.096	0.097
Δ SFT	0.004	0.003	0.004

1.3 Supplementary Table S3: Format-Matched Routing Control

Table 3: Layer-averaged JS divergence before and after format matching.

Condition	JS Divergence	Retention Ratio
Original (heterogeneous formats)	0.160	100%
Format-matched (neutral template)	0.145	90.2%

1.4 Supplementary Table S4: Head-to-Head Comparison Against Classical and PLM Baselines

Table 4: Remote homology accuracy on identical 500-pair balanced evaluation (`protein_pair_remote`, seed 42). All methods evaluated on the same pair set used by v4 / v5. Classical alignment tools (MMseqs2, DIAMOND) operate on raw sequence; PLM baselines use mean-pooled last-layer embeddings with cosine similarity at threshold 0.5.

Method	Type	Acc (default)	Acc (optimal)	RO
ESM-2 650M	PLM (mean-cosine)	50.50%	51.80%	
ESM-2 3B	PLM (mean-cosine)	50.00%	51.20%	
MMseqs2 (easy-search, s=7.5)	alignment	54.40%	55.60%	
DIAMOND (more-sensitive)	alignment	53.20%	53.20%	
Gemma-4-Instruct (vocab-extended, zero-shot)	LM (chat)	60.00%	—	
OmniGene-4 v3	LM (chat)	59.50%	—	
OmniGene-4 v4	LM (chat)	82.00%	—	
OmniGene-4 v5	LM (chat, dual-head)	82.60%	—	

The ESM-2 family does not improve materially when scaled from 650M to 3B parameters (+0.7 pp), confirming that the bottleneck on this remote-homology distribution is not encoder capacity but rather the absence of explicit homology supervision during pretraining. Classical alignment tools (MMseqs2, DIAMOND) reach 53–56%, marginally above chance, indicating that the 500 pairs in this sample are genuinely remote in sequence space (no detectable sequence alignment under default E-value cutoffs in >95% of pairs). OmniGene-4 v4 / v5 reach 82% on the same pairs, with the gap attributable to the v4 SFT-recipe changes (Alpaca template, loss masking, task oversampling).

1.5 Supplementary Table S8: Case-Study Pair Details

Table 5: Five case-study protein pairs drawn from the 500-pair Remote sample. ESM2-650 / 3B columns report binary prediction at cosine threshold 0.5; MMseqs / DIAMOND columns are 1 if a hit was found at $E \leq 10^{-3}$, 0 otherwise. v5 column is the parsed answer of OmniGene-4 v5. Type-A = MMseqs/DIAMOND fail but v5 succeeds; Type-B = ESM-2 false positive but v5 correctly rejects; Type-C = sanity check (all correct); Type-D = limitation (v5 fails).

Case	Type	Len 1	Len 2	Label	ESM2-650	ESM2-3B	MMseqs	DIAMOND	v5
1	A	107	106	1	1 (Y)	1 (Y)	0 (N)	0 (N)	1 (Y)
2	A	142	161	1	1 (Y)	1 (Y)	0 (N)	0 (N)	1 (Y)
3	B	244	157	0	1 (N)	1 (N)	0 (Y)	0 (Y)	0 (Y)
4	C	143	101	1	1 (Y)	1 (Y)	1 (Y)	1 (Y)	1 (Y)
5	D	108	176	1	1 (Y)	1 (Y)	0 (N)	0 (N)	0 (N)

1.6 Supplementary Table S9: Layer-12 Top-3 Routed Experts per Case

Table 6: Top-3 routed experts at layer 12 for each case-study pair. The same expert set $\{E_{50}, E_9, E_{108}\}$ activates on all five cases, supporting the conclusion that homology decisions occur inside expert computation rather than at the routing gate (Section 3.5).

Case	Outcome	Top-1	Top-2	Top-3
1	v5 correct (label=1)	E50	E9	E39
2	v5 correct (label=1)	E50	E9	E108
3	v5 correct (label=0)	E50	E9	E108
4	v5 correct (label=1)	E50	E108	E9
5	v5 <i>wrong</i> (label=1)	E50	E108	E9

Layer-12 entropy is also stable across cases ($H = 2.70\text{--}2.83$), and per-layer JS divergence between any homologous case and the non-homologous Case 3 stays below 0.04 across all 30 layers (peak at layer 5, 0.040). For comparison, the layer-averaged cross-task JS reported in the main paper is 0.230 (see Table S1 / Figure S1). The difference of more than $5\times$ between cross-case and cross-task JS supports the gate-vs-expert interpretation in Section 3.5.

2 Supplementary Figures

2.1 Supplementary Figure S1: Per-Layer JS Curves

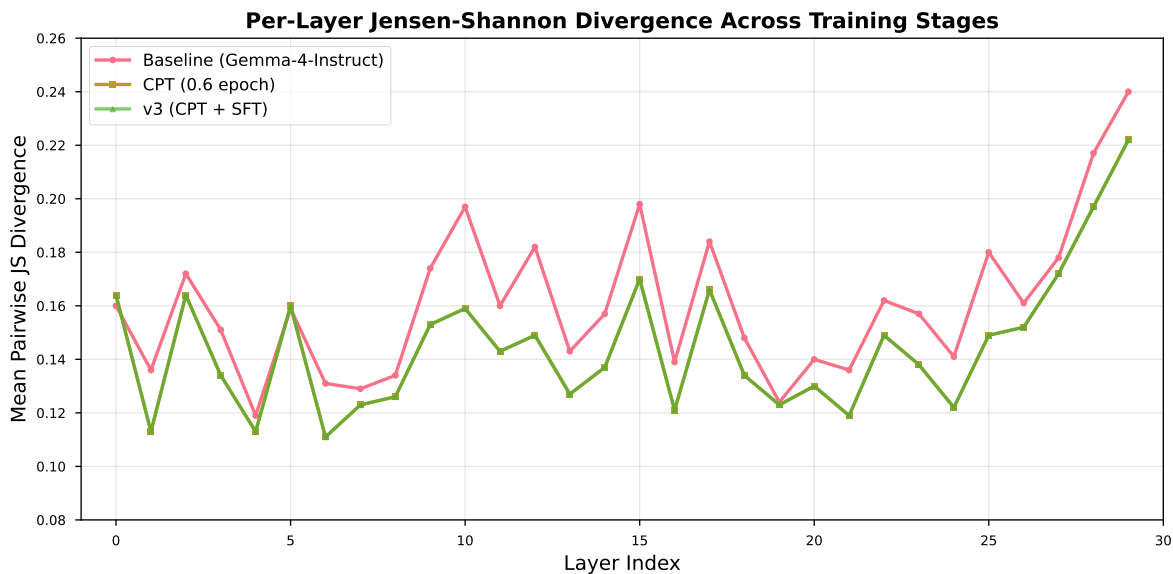


Figure 1: Layer-by-layer Jensen-Shannon divergence across training stages. The CPT stage shows the largest increase in middle layers (L11–L22), while the SFT stage shows concentrated changes near the output layers (L28–L29).

2.2 Supplementary Figure S2: Expert Activation Heatmap

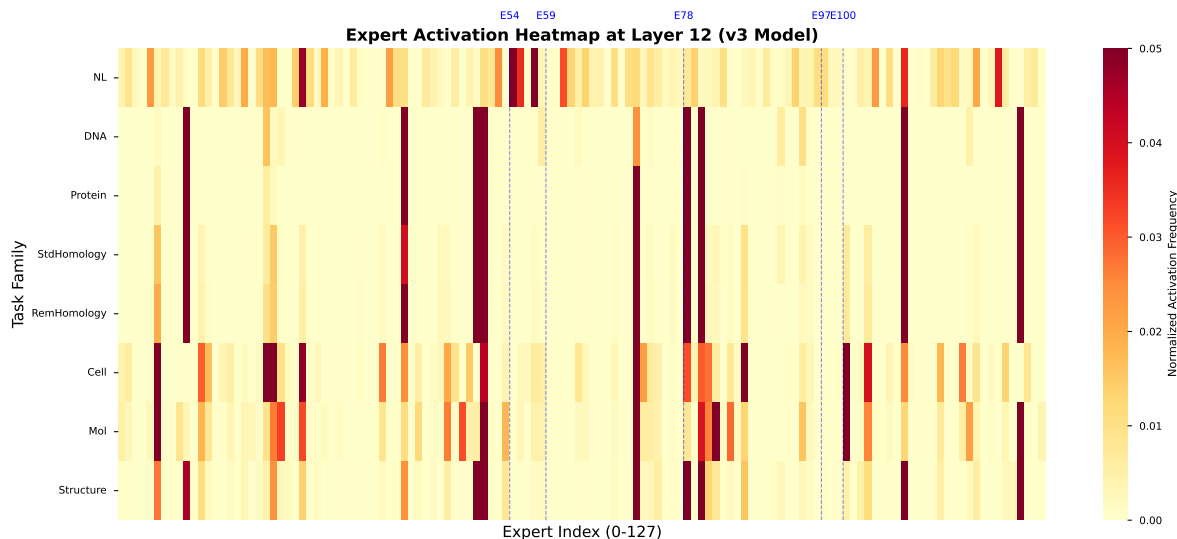


Figure 2: Expert activation heatmap at Layer 12 for eight task families. Rows: tasks (NL, DNA, Protein, StdHom, RemHom, Cell, Mol, Structure). Columns: 128 experts. Color intensity indicates normalized activation frequency. Blue dashed lines mark key experts (E54: English function words, E59: DNA 2-mers, E78: amino acids, E97: cell markers, E100: SMILES punctuation).

2.3 Supplementary Figure S3: Bootstrap Distribution

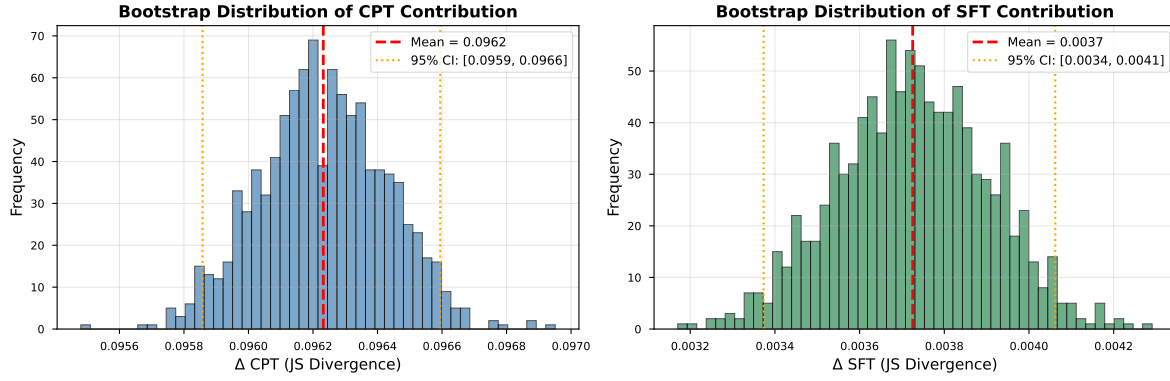


Figure 3: Bootstrap distribution of CPT and SFT contributions to JS divergence (1,000 iterations). Both distributions are narrow and exclude zero, confirming statistical robustness. Left: Δ CPT = 0.096 [0.096, 0.097]. Right: Δ SFT = 0.004 [0.003, 0.004].

3 Supplementary Methods

3.1 Detailed Training Hyperparameters

3.1.1 Continued Pretraining (CPT)

Table 7: CPT hyperparameters for OmniGene-4 v3.

Parameter	Value
Base model	Gemma-4-26B-A4B-Instruct
Vocabulary size	290,048 (262,020 + 28,028 bio tokens)
Training data	32.5 GB (8.73B tokens)
Batch size	6 per GPU $\times$ 8 GPUs $\times$ 4 accum = 192
Learning rate	2e-5
LR scheduler	Cosine with 3% warmup
Optimizer	paged_adamw_8bit
Weight decay	0.01
Max gradient norm	1.0
Sequence length	1024
Epochs	0.6
Total steps	2,806
LoRA rank	64
LoRA alpha	128
LoRA dropout	0.05
LoRA targets	q/k/v/o_proj, gate/up/down_proj, router.proj
Precision	bfloat16
Gradient checkpointing	True

3.1.2 Supervised Fine-Tuning (SFT)

Table 8: SFT hyperparameters across all five stages. v2/v3 used a chat-style template; v4 switched to pure Alpaca with prompt-token loss masking and task-level oversampling (Structure x3, Mutation x2). v5 adds a dual-head architecture (3Di + DSSP per-residue classifiers) trained jointly with the generation head under a 0.5 / 0.5 loss split, on the Structure-only subset.

Parameter	v2 / v3	v4	v5
Init from	v2 / Gemma-bio	v3 LoRA + embed	v4 LoRA + embed
Training data	179K / 199K instr	199K (resampled)	21K Structure subset
Prompt template	chat tags	pure Alpaca	pure Alpaca
Loss masking on prompt	no	yes (-100)	yes (-100)
Joint loss	gen CE only	gen CE only	0.5 gen + 0.5 cls
Per-device batch size	4	2	2
Gradient accumulation	16	32	16
Effective batch size	64	64	32
Learning rate	5e-5 / 2e-5	2e-5	1e-5
LR scheduler	cosine, 3% warm	cosine, 3% warm	cosine, 5% warm
Optimizer	paged_adamw_8bit	paged_adamw_8bit	paged_adamw_8bit
Sequence length	1024	1536	1536
Epochs / Steps	1 ep / 3,118	/ 4,257	2 ep / 1,350
LoRA rank, alpha	64, 128	64, 128	64, 128
LoRA dropout	0.05	0.05	0.05
GPU-hours (single H20)	11.8 / 13.2	30	5
Precision	bfloat16	bfloat16	bfloat16

3.2 Data Mixture Composition

3.2.1 CPT Data Mixture

Table 9: CPT data mixture composition (32.5 GB total).

Source	Size (GB)	Tokens (B)	Proportion
DNA (human genome)	8.0	2.1	24.6%
Protein (UniProt)	8.0	2.1	24.6%
Protein (LucaOne)	7.5	2.0	23.1%
OpenWebText	8.0	2.1	24.6%
Structure (3Di + DSSP)	0.4	0.1	1.2%
Instruction replay	0.6	0.4	1.9%
Total	32.5	8.7	100%

3.2.2 SFT Data Mixture

Table 10: SFT data mixture composition (199,576 examples).

Task Family	Examples	Proportion
Protein homology	49,894	25.0%
Literature (UniProtQA)	39,915	20.0%
Mutation (MutaDescribe)	29,936	15.0%
Cell biology	29,936	15.0%
Molecule (SMILES)	25,945	13.0%
Structure (3D)	19,958	10.0%
DNA homology	3,992	2.0%
Total	199,576	100%

3.3 Routing Collection Protocol

3.3.1 Forward Hook Implementation

Listing 1: Forward hook for collecting router activations

```
def make_hook(layer_idx):
    def hook(module, input, output):
        # output = (router_probs, top_k_weights, top_k_indices)
        # top_k_indices shape: [seq_len, 8]
        top_k_indices = output[2]
        for seq_idx in range(top_k_indices.shape[0]):
            for k_idx in range(top_k_indices.shape[1]):
                expert_id = int(top_k_indices[seq_idx, k_idx].item())
                routing_counts[current_task][layer_idx, expert_id] += 1
    return hook

# Register hooks on all layers
handles = []
for i in range(NUM_LAYERS):
    h = model.model.language_model.layers[i].router.register_forward_hook(
        make_hook(i))
    handles.append(h)
```

3.3.2 Bootstrap Resampling Protocol

Listing 2: Prompt-level bootstrap resampling

```
def bootstrap_sample(prompt_counts_dict, tasks):
    """
    prompt_counts_dict: {task: [n_prompts, layers, experts]}
    Returns: resampled counts_dict {task: [layers, experts]}
    """
    resampled = {}
    for task in tasks:
        prompts = prompt_counts_dict[task]
```

```

        n_prompts = prompts.shape[0]
        # Resample with replacement
        indices = np.random.choice(n_prompts, n_prompts, replace=True)
        resampled_prompts = prompts[indices]
        # Sum over prompts
        resampled[task] = resampled_prompts.sum(axis=0)
    return resampled

# Run 1000 bootstrap iterations
for i in range(1000):
    baseline_boot = bootstrap_sample(baseline_prompts, tasks)
    cpt_boot = bootstrap_sample(cpt_prompts, tasks)
    v3_boot = bootstrap_sample(v3_prompts, tasks)
    # Compute JS and store
    ...

```

4 Supplementary Results

4.1 Token-Level Expert Specialization

Table 11: Top-5 tokens for selected experts at Layer 12 (v3 model).

Expert	Purity	Top-5 Tokens
E54	80%	the, of, and, to, in
E59	46%	AA, AT, TT, GG, CC
E78	35%	M, L, V, I, F
E97	28%	CD4, CD8, IL, TNF, IFN
E100	15%	(,), =, C, 0

4.2 Per-Task Entropy Analysis

Table 12: Shannon entropy of expert distributions across training stages.

Task	Baseline	CPT	v3 (CPT+SFT)
NL	3.85	4.13	4.15
DNA	2.89	2.76	2.78
Protein	2.76	2.54	2.56
StdHomology	2.68	2.48	2.50
RemHomology	2.71	2.52	2.54
Cell	2.95	2.88	2.90
Mol	3.02	2.95	2.97
Structure	2.82	2.65	2.67

4.3 Supplementary Table S5: Multi-task Generation Evaluation (v5)

Table 13: Multi-task generation accuracy across six evaluation categories, 100 samples each, on a held-out split. Keyword-overlap is the standard text-overlap score; Structure additionally reports character-level overlap on the 3Di / DSSP output strings, which are not natural-language tokens. The low Mutation and Literature scores reflect known long-tail and abstract-language difficulties; the dual-head architecture in v5 specifically addresses Structure via the classification heads (Table S6) rather than via generation.

Category	Keyword score	Char overlap
Protein	1.000	—
Mol	0.980	—
Cell	0.966	—
Literature	0.559	—
Mutation	0.113	—
Structure (gen mode)	0.000	0.259

4.4 Supplementary Table S6: Per-residue Classification Head Accuracy (v5)

Table 14: Per-residue accuracy of the two classification heads on a 50-protein held-out evaluation, under teacher-forced reference-residue inference. Scores are aggregated by averaging per-protein accuracy. Both heads operate on the same final-hidden-state representation as the generation head; the dual-head architecture trains them jointly with the generation objective ($\alpha = \beta = 0.5$) for 1,350 steps.

Head	Classes	Chance	Accuracy	N proteins
3Di (Foldseek)	20	5.0%	78.6%	19
DSSP (secondary str.)	8	12.5%	100.0%	31

4.5 Supplementary Table S7: Standard / Remote Homology Updated Numbers

Table 15: BioPAWS homology accuracy across all five stages on consistent evaluation samples. Standard uses 1,000 balanced pairs (`protein_pair_short`); Remote uses 500 balanced pairs (`protein_pair_remote`). The v3-to-v4 jump is the headline finding of the v4 recipe change (Alpaca + loss masking + task oversampling).

Stage	Standard	Remote	BixBench
Gemma-4-Instruct (vocab-extended)	85.0%	60.0%	87.0%
v2 (CPT + Bio-SFT)	100.0%	56.6%	91.0%
v3 (+20K remote pairs)	99.95%	59.5%	93.7%
v4 (Alpaca + loss-mask + oversamp)	99.5%	82.0%	90.4%
v5 (+ dual-head)	99.4%	82.6%	93.7%

5 Code Availability

All code, training scripts, and analysis notebooks are available at:

<https://github.com/maris205/omnigene4>

5.1 Key Scripts

- `biopaws/cpt/1-prepare_cpt_data_mp.py` – CPT data preparation (multi-process)
- `biopaws/cpt/2-run_cpt.py` – CPT training (8-GPU DDP)
- `biopaws/sft_data/17-train_bio_sft_v3_remote.py` – SFT v3 training
- `biopaws/cpt/40-train_bio_sft_v4.py` – SFT v4 (Alpaca + loss-mask + oversampling)
- `biopaws/cpt/41-train_bio_sft_v5_classifier.py` – SFT v5 (dual-head)
- `biopaws/cpt/18-eval_v3_sft.py` – v3 evaluation
- `biopaws/cpt/42-eval_v4_full.py` – v4 evaluation
- `biopaws/cpt/43-eval_v5_full.py` – v5 evaluation (incl. classification heads)
- `biopaws/cpt/44-merge_v5_to_full.py` – v5 LoRA + heads -> BF16 merge
- `biopaws/cpt/20-collect_moe_activations.py` – Routing collection
- `biopaws/cpt/26-format_matched_routing.py` – Format control
- `biopaws/cpt/27-esm2_remote_headtohead.py` – ESM-2 comparison
- `biopaws/cpt/28-bootstrap_ci.py` – Bootstrap CI

5.2 Data Files

Routing-count matrices and analysis results are in `outputs/moe_analysis/`:

- `moe_counts_baseline.npz` – Baseline routing (50 KB)
- `moe_counts_cpt.npz` – CPT routing (47 KB)
- `moe_counts_v3.npz` – v3 routing (48 KB)
- `format_matched_report.json` – Format control results
- `esm2_remote_2k_eval.json` – ESM-2 comparison results
- `bootstrap_ci_report.json` – Bootstrap CI results
- `bootstrap_samples.npz` – Full bootstrap samples (1,000 iterations)
- `v4_full_eval.json`, `v5_full_eval.json` – v4 / v5 multi-task and classification-head accuracy

5.3 Model Weights

All model artefacts (LoRA adapters, classification heads, merged BF16 weights, tokenizer, chat template) are publicly released on the Hugging Face Hub under the **dnagpt** organization:

- <https://huggingface.co/dnagpt/OmniGene-4-CPT-v2-merged> – Base + CPT only, BF16 (50 GB)
- <https://huggingface.co/dnagpt/OmniGene-4-SFT-v3-merged> – CPT + SFT v3, BF16 (50 GB)
- <https://huggingface.co/dnagpt/OmniGene-4-SFT-v3-GGUF> – v3 quantized GGUF Q4_K_M (16 GB)
- <https://huggingface.co/dnagpt/OmniGene-4-SFT-v4> – v4 LoRA + embedding (1.9 GB)
- <https://huggingface.co/dnagpt/OmniGene-4-SFT-v5> – v5 LoRA + embedding + classification heads (1.9 GB)
- <https://huggingface.co/dnagpt/OmniGene-4-SFT-v5-merged> – v5 final, BF16 with classification heads (52 GB, 11 shards)